

ROHAN KOLHE

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Profile

As a Computation and Neural Systems major at the California Institute of Technology, I have developed expertise across computer science, applied mathematics, statistics, neuroscience, and electrical engineering. My work spans from complex analysis to neurobiology, applying advanced AI techniques to real-world problems. I am passionate about developing machine learning models, neural network architectures, and data-driven approaches to enhance AI applications in neuroscience and computational biology.

Research & Publications

Batik: A Multimodal Large Language Model-Based System for End-to-End Social Behavior Discovery, Interpretation, and Annotation

Rohan Kolhe†, Nestor Coria†, Jadon Hale†, Jineun Kim, Amit Vinograd, Dan Biderman, Kelly Buchanan, Pietro Perona, Scott Linderman, David Anderson, Aditya Nair

In Review at Nature Methods

Experience

Undergraduate Research Fellow – Anderson Lab, Caltech

Feb. 2024 – Present

- Apply vision transformers to annotate mouse behavior videos, achieving high-accuracy classification for behaviors such as attack and mounting.
- Integrate multimodal large language models with clustering methods to identify and interpret novel behavioral patterns.
- Investigate transformer model layers to explore their neural network-like activity in behavioral analysis.

Undergraduate Research Fellow – Shapiro Lab, Caltech

June 2023 – Aug. 2023

- Developed an algorithm using the Finite Element Method (FEM) and FEniCSX to predict brain tissue properties through viscoelastic analysis of MRI-generated strain fields.
- Modeled mechanical properties of brain structures to enhance understanding of tissue differentiation in medical imaging.

Undergraduate Research Fellow – Shapiro Lab, Caltech

June 2022 – Aug. 2022

- Modeled gas vesicle protein structures subjected to ultrasound pressure to study the buckling effect.
- Simulated material and geometric imperfections to analyze their impact on vesicle stability and mechanical response.

Projects

Classifying Pathologies Through Chest X-Ray Images (2024)

- Applied generative adversarial networks (GANs) and convolutional neural networks (CNNs) such as ResNet and DenseNet to predict pathologies from the CheXpert dataset.

- Achieved competitive performance, placing third in the competition.

Analysis of Reinforcement Agent Signaling Strategies in Heads-Up Poker (2024)

- Explored signaling strategies developed by Bayesian and Deep Q-Network (DQN) agents in a multi-agent reinforcement learning environment.
- Analyzed agent behavior using game theory concepts such as the theory of mind.

Uncertainty Quantification on Protein Fitness (2023)

- Implemented Bayesian Neural Networks using Monte Carlo Markov Chains and Stochastic Variational Inference to optimize protein fitness predictions.
- Explored conformal prediction, Gaussian processes, and Bayesian modeling to enhance model robustness.

Education

Bachelor of Science – Computation and Neural Systems

California Institute of Technology — Expected 2025

Relevant Coursework:

Uncertainty Quantification in ML — Decidability and Tractability — Data Structures — ML and Data Mining — Learning Systems — Software Design — Waves and Quantum Mechanics — Probability Models — Complex and Real Analysis — Signal Processing — Behavioral Game Theory — Statistical Inference

Leadership & Extracurriculars

NeuroSculpt – Founder & Treasurer (Feb. 2024 – Present)

- Established the NeurotechX chapter at Caltech to educate students about Brain-Machine Interfaces (BMI).
- Lead projects in naturalistic EMG decoding and participate in regional competitions.

Peer Academic Coach – Caltech (Jan. 2022 – Present)

- Tutor Caltech students in STEM subjects, helping them develop independent learning strategies.

CalTutors – Tutor (Sep. 2024 – Present)

- Provide tutoring in physics, chemistry, biology, mathematics, test preparation, and computer science to high school students.

Awards & Skills

- **Technical Skills:** Python, Java, Angular, C
- **Languages:** Marathi, Spanish
- **Honors:** Bonsall SURF Technical Writing Prize (2025), Named Undergraduate Research Fellow (2022, 2023, 2024)